

Q.1

a. Process Planning

Process Planning is a act to determine a appropriate manufacturing Process & Sequence of operation to Convert raw material into finished Product.

b. Hybrid CAPP

- Hybrid CAPP is Combination of both Variant & Generative CAPP
- Initially it start with standard plan (Variant) & further utilizes (Generative) Automation decision logic, rule & Algorithm
- It is a ~~Pr~~ flexible & Practical approach Combination

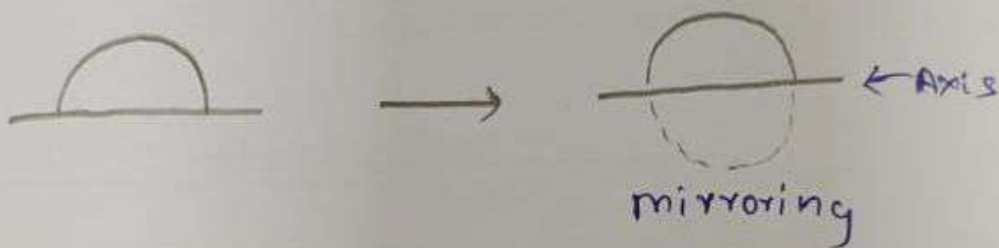
c. Meshing in FEA

- Meshing in FEA is done to Convert large, Complex geometry into Small finite element.
- Element are connected with nodes where equation are solved
- To Increase Accuracy, meshing is used.

d → $3 + 7 * 2$

output = 17

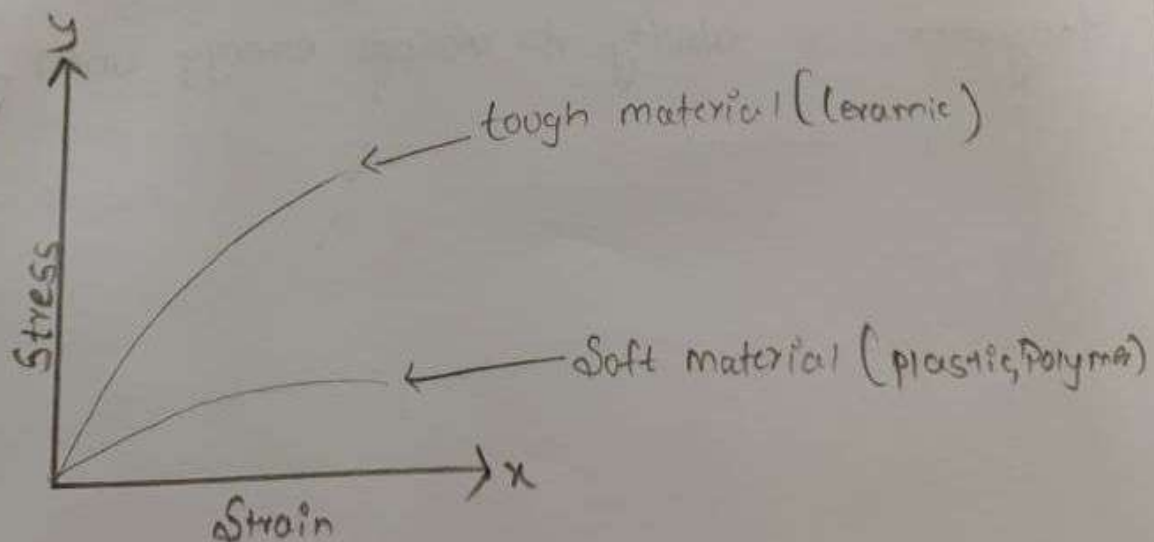
e → mirror tool Copy the structure around the axis



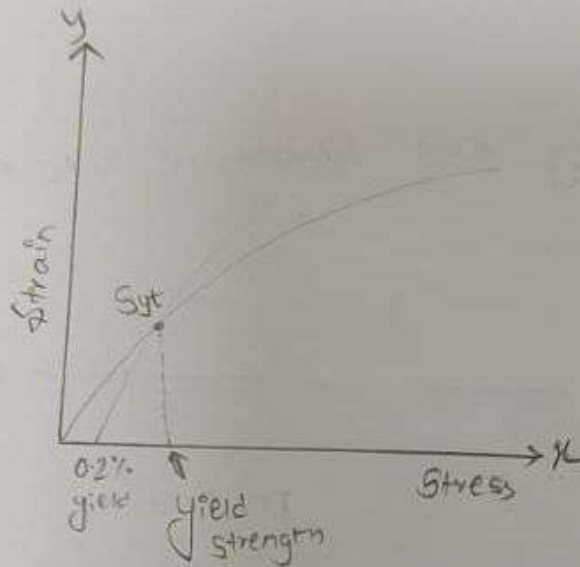
Mirroring is used in Design Software to reduce the work, by making one side of structure it is mirror another side.

d → clc Command Clears Variable from workspace

g.



h $\rightarrow$ 0.2% Offset is used to determine yield strength when the yield point is not clearly visible in stress-strain graph, 0.2% offset yield method is used to determine yield strength



i $\rightarrow$ Toughness

It is area under stress strain curve represent ability to absorb energy per unit volume of material is called as Toughness

toughness is ability to absorb energy until failure

J → G & M Codes

G → Geometric Codes

It control the movement of CNC machine while lathing, milling, cutting

Example - G00 - Rapid positioning

G01 - Linear movement

G02 - Circular interpolation

M → Miscellaneous Code

It control the function of CNC machine such as Spindle, Coolant, etc.

Example - M04 - Spindle ON

M05 - Spindle OFF

K. CNC → Computer Numerical Control

L → 2D Element are used for modelling thin structures

Example - thin disc

- Patch

Q.3 CAPP (Computer Aided Process Planning)

- CAPP is Computer integration in Process Planning
- CAPP is bridge between CAD & CAM use for Automation & optimization of Process

Process Planning

Process planning is act of determining the appropriate manufacturing Process & Sequence of operation to convert raw material into finished Product

Advantages of CAPP

1) Automation of Process

- Reduce manual Effort & Human dependency in Process Planning

2) Improve Consistency

- Generate Standard Process Plan to Avoid Planner to Planner Variation

3) Reduced lead time

Rapid generation of Process plan &
Shortened Production time

4) Cost optimization

° ~~Optimize~~ Optimize Process, Machine, tool
to reduce Cost of manufacturing

5) Knowledge retention

° Capture Expert Process in a digital
System & Preserve expertise.

6) Flexibility

Easy to adapt a new design for a Part
in families

Approaches in CAPP.

① Variant CAPP

• Variant CAPP based on Group technology
Concept.

- Such that Part with similar design & manufacturing characteristic are grouped in similar part ~~in~~ families
- Process plans are stored for each families
- For a new part, the existing plan with similar part are modified & optimized

Steps for Variant CAPP

- Classify Part into family using coding system
- Retrieval of standard process for part families
- Edit & modify for addition & optimize

Advantages

- Simple & Easy for implementation
- Utilize existing knowledge
- Suitable for batch Production

Limitation

- It is not fully automated
- Initial Classification of Part families
- Limited adaption

② Generative CAPP

- Generative CAPP based on Knowledge base or A.I technique for operation, machine, tools
- Process plan are Automatically generated by using decision logic, rule, Algorithm
- The new Part is analyse & Product, tolerance material Properties are opted to form a new Part from Scratch.

Advantages

- It is fully automated method
- It can be integrated With Computer technology,

Limitation

- It require large knowledge data base
- It is costly & tedious.

Q4. CNC Programming

CNC Programming is a Computer Numerical Control Program use for written instruction to be given to CNC machine for cutting, milling, lathe Process.

Code in CNC Programming

- Geometric Code (G-Code)

1) Control the movement of CNC machine

Ex. G00 → Rapid Positioning

G01 → Linear movement

G02 → Circular interpolation

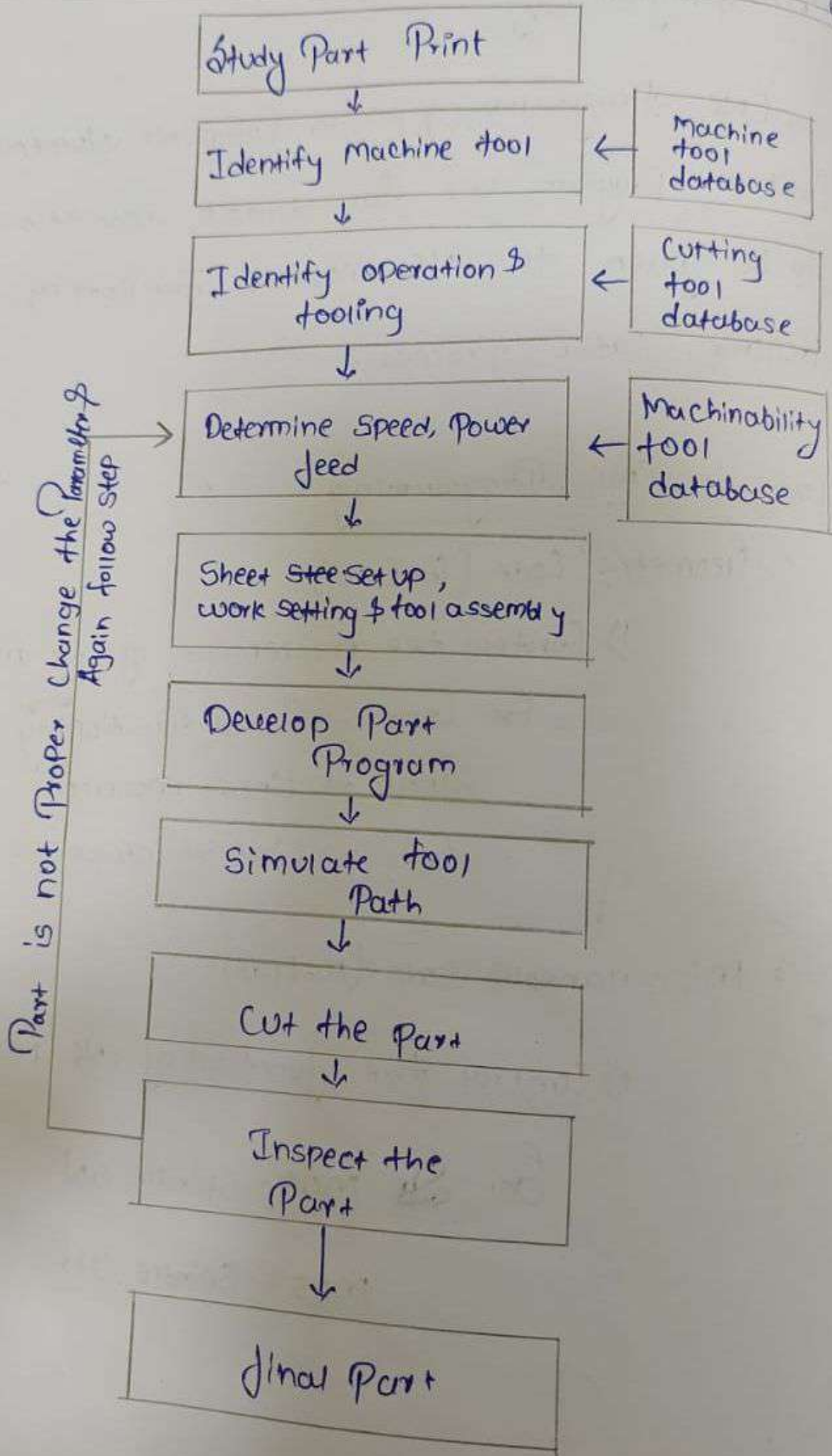
- Miscellaneous Code (M-Code)

1) Control the function of CNC machine

Ex. ~~S~~ M04 - Spindle ON

M05 - Spindle OFF.

Steps involve in CAD-CAM → CNC Part Programming



Process:-

- First the design of Part is created using CAD (Computer Aided design) Software
- After designing, Suitable machine, tool are opted for Part
- After machine Selection, different Parameter are Selected according to Part such as Power, Feed Speed using machinability tool data base
- Part Program is developed to Simulate tool Path
- Sheet is setup on which Part is fabricated
- Cut the Part using CNC machine & Inspect the Part
- If Part is not according to design or Changes required.
- Change the Parameter in machinability tool database & again follow the steps
- Finally the Part is obtained.

Q.5 Finite Element Analysis

Finite Element Analysis is a Computer based numerical technology use to predict how the structure or material behave under external forces such as

- Heat
- Pressure
- Deformation

Instead of solving complex equation for large object, object is divided into smaller & simple part called Element

- The Computer calculate the behaviour of element while expose to external force, all the ~~be~~ behaviour of element are combine to get the final result of object.

Application

- Stress analysis of microneedle & Insulin Pen
- Bone & tissue deformation due to Implants.

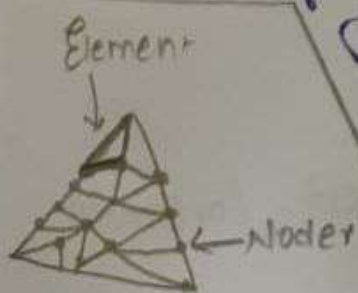
Steps of Finite Element Analysis

1) Geometry Creation

- Using Cad Software, the 3d design is Prepared on which the Analysis is done

2) Meshing

- Also known as discretization
- In meshing the large & complex geometry is converted to small finite element
- The element are connected through a nodes where equation are solved.
- ~~Convergent~~ Convergence test: It is done to find the optimum number of element require for accuracy
- If the difference between two meshing is less than 5% it can be called Converged



• For improving Accuracy

- Increase number of elements
- Increase order of elements

3. Element type

Type	Structure	Application	Example
1) 1D Element	Line	Simple load transfer on beam or frame	• modeling Stent, Catheter wire
2) 2D Element	Triangle Quadrilateral	Thin structure like shell, membrane, wall.	Thin Disc, Patch
3) 3D Element	Tetrahedron Hexahedron	Solid structure with filled volume	Implants

4. Application of material

The material Property Such as mechanical or thermal are studied using Young's modulus, Poisson's ratio & thermal conductivity

5. Boundary Conditioning

The external force Such as heat, Constraint Pressure are applied.

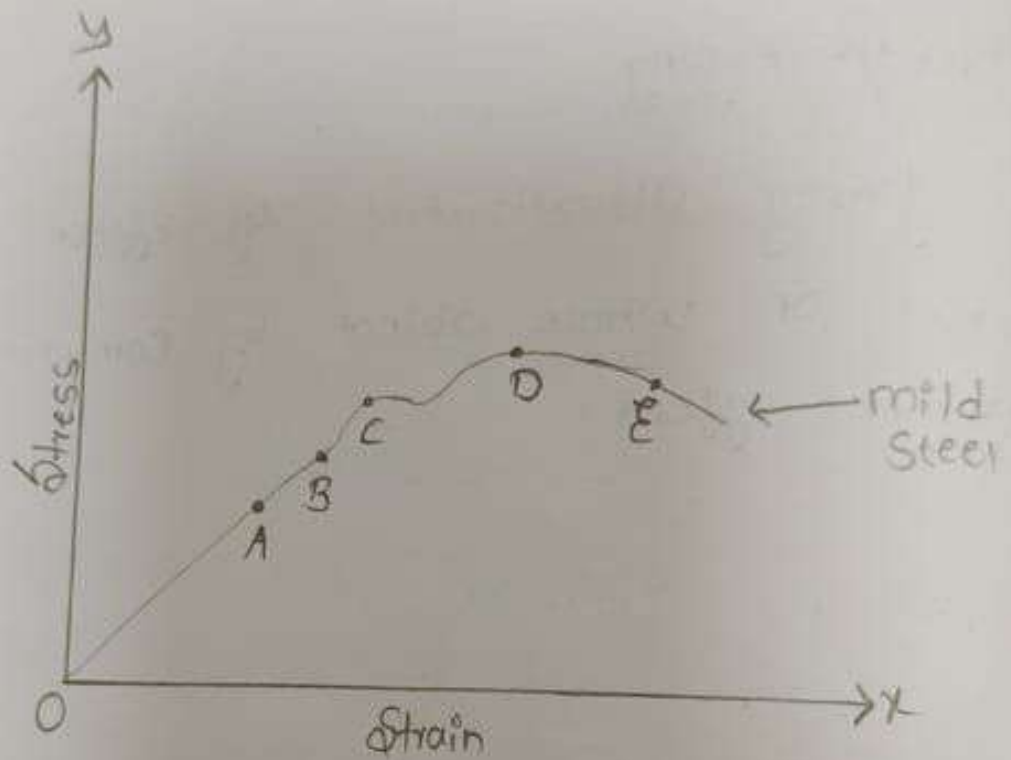
6. Solving

The computer calculate the effect of external forces on each element

7. Post Processing

finally Visualization of Effect of external forces on whole object by combining all element effect.

Q 6.1) Stress Strain Curve demonstrate how a material react to increase tensile strength, Constrain & deformation



O $\rightarrow$ Starting Point of test

A $\rightarrow$ Proportionality limit

B $\rightarrow$ Elastic limit

C $\rightarrow$ Yield Point

D $\rightarrow$ Ultimate tensile Strength

E $\rightarrow$ Fracture Point

0 $\rightarrow$ It is origin or starting point of graph.

Stress $\rightarrow$ Force applied Per unit area

Strain $\rightarrow$ deformation occur Per unit area

Point A :- Proportionality limit

- ~~It is directly Prop~~
- Point A describe that Stress is directly Proportional to Strain, it obey Hooke's law
- It show Perfectly Elasticity

Point B :- Elastic limit

- Upto this limit, material can come to it's original shape after unloading of stress
- Deformation is still Elastic in this region.

Point C : Yield Point

- It shows the maximum stress up to which material shows elastic behaviour.
- Yield Point in stress-strain curve represents the limit to elastic behaviour & the starting of plastic behaviour.

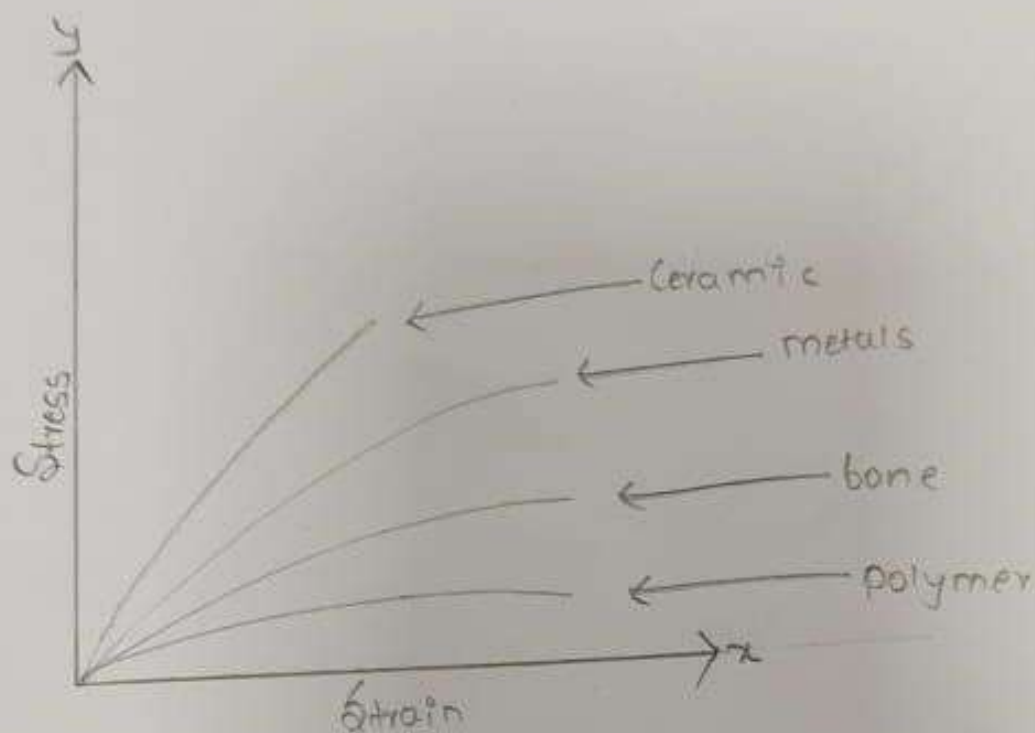
Point D : Ultimate Tensile Strength

- Point D represents the maximum stress the material can withstand.
- After this it starts necking.

Point E : Fracture Point

- At this point the material finally breaks.

Q.6 ii) Stress - Strain diagram of Ceramic, Polymer, bone & metals.



Stress strain graph of different material

- ① Ceramic $\rightarrow$ Brittle in nature
- ② Metal & bone $\rightarrow$ Ductile in nature
- ③ Polymer $\rightarrow$ Plastic nature